

the other (Blackwell, 1953).

Assume that the human's objective is to match the action to the state. Thus, the human faces a classification problem. It is without loss of generality to normalize the payoff from taking the action that matches the state to zero. Let $c_{FP,r}$ be the disutility of human r if she takes the action $a = 1$ when the state is $\omega = 0$ (false positive) and $c_{FN,r}$ be the disutility if she takes the action $a = 0$ when the state is $\omega = 1$ (false negative). The payoff of the human is therefore

$$u_r(a, \omega) = -1 \cdot \{a = 1, \omega = 0\} \cdot c_{FP,r} - 1 \cdot \{a = 0, \omega = 1\} \cdot c_{FN,r}. \quad (1)$$

We allow the human's posterior belief given the observed signals to deviate from those implied by the true probability law $\pi_r(\cdot|\omega)$. Specifically, let $s_{ir} \subset \{s_r^A, s_{ir}^E\}$ be the subset of signal realizations observed by the human r and $p_r(\omega | s_{ir}) \in [0, 1]$ be the human's posterior belief that the state is $\omega \in \{0, 1\}$ when she observes s_{ir} . Suppressing the dependence of signals on the pair (i, r) , the human's action given the signal s is

$$a_r^*(s; p_r) = 1 \cdot \left\{ \frac{p_r(\omega = 1 | s)}{p_r(\omega = 0 | s)} > c_{rel,r} \equiv \frac{c_{FP,r}}{c_{FN,r}} \right\}. \quad (2)$$

The expected payoff from following $a^*(s)$ is

$$V_r(s; p_r) = E[u_r(a_r^*(s; p_r), \omega) | s] = \sum_{\omega} u(a_r^*(s; p_r), \omega) \pi_r(\omega | s),$$

where decisions are based on the human's belief p_r , but are evaluated according to the true law π_r . Because we allow for p_r to differ from π_r , the odds ratio $\frac{p_r(\omega=1|s)}{p_r(\omega=0|s)}$ may differ from the analogous quantity constructed using π_r . Thus, the action $a_r^*(s; p_r)$ can deviate from the optimal action $a_r^*(s; \pi_r)$ given the signal $s = (s^A, s^E)$. Except in knife-edge cases, $V_r(s; p_r)$ is lower than $V(s; \pi_r)$ whenever $a^*(s; p_r) \neq a^*(s; \pi_r)$.

A key objective is to analyze deviations in humans' use of AI signals from the benchmark model of Bayesian updating with correct beliefs (about the joint distribution of the signals and the state). Bayes' rule implies that, given the signals (s_i^A, s_{ir}^E) , the log-odds of $\omega = 1$ to $\omega = 0$ —the key decision-relevant quantity—is given by

$$\log \frac{\pi_r(\omega_i = 1 | s_i^A, s_{ir}^E)}{\pi_r(\omega_i = 0 | s_i^A, s_{ir}^E)} = \log \frac{\pi_r(s_i^A | \omega_i = 1, s_{ir}^E)}{\pi_r(s_i^A | \omega_i = 0, s_{ir}^E)} + \log \frac{\pi_r(\omega_i = 1 | s_{ir}^E)}{\pi_r(\omega_i = 0 | s_{ir}^E)}. \quad (3)$$

The second term on the right-hand side is the posterior log-odds ratio for the two states

$\omega_i = 1$ to $\omega_i = 0$ given that the human’s signal is s_{ir}^E . The first term is positive if, given the realization s_{ir}^E , the signal s_i^A is more likely if $\omega_i = 1$ as compared to $\omega_i = 0$. In this case, the posterior odds shift in favor of the state $\omega_i = 1$. Analogously, the odds shift away from $\omega_i = 1$ if, given s_{ir}^E , the signal s_i^A is more likely if the state were $\omega_i = 0$.

The task of empirically analyzing deviations in participants’ beliefs from the benchmark in equation (3) is challenging because the signals differ across cases i and humans may, due to differences in skill, have heterogeneous signal distributions $\pi_r(\cdot)$. The ideal dataset would elicit $p_r(\omega_i = 1 \mid s_i^A, s_{ir}^E)$ and $\pi_r(\omega_i = 1 \mid s_{ir}^E)$ for the same case and the same human to keep the signals (s_i^A, s_{ir}^E) , and human-specific parameters fixed, and use the latter to estimate $\pi_r(\omega_i = 1 \mid s_i^A, s_{ir}^E)$ using equation (3).

However, empirically implementing this strategy requires us to confront several experimental and methodological challenges. First, the experiment needs to be sensitive to anchoring and order effects when eliciting $p_r(\omega_i = 1 \mid s_i^A, s_{ir}^E)$ and $\pi_r(\omega_i = 1 \mid s_{ir}^E)$ from participants. Second, the potential for measurement error and the dependence of the first term in equation (3) on a latent signal s_{ir}^E complicates the exercise. We defer our approach to this second challenge to section 5, and turn our attention to the experimental design.

3 Setting and Experiment

Our experiment elicits the probability of a pathology’s presence $p_r(\omega = 1 \mid s)$ and a recommended treatment/follow-up decision a under varying information treatments. There are four information treatments in the experiment, with only the chest X-ray in the minimal information case, to which we add AI assistance, contextual information, or both. Next, we describe the experimental context and interface before presenting the design of our experiments.

3.1 Experimental Context

3.1.1 Radiology

Radiologists diagnose the presence of a given pathology at the request of a treating physician. The information available to a radiologist consists of diagnostic images (e.g. chest X-rays), any relevant medical history (e.g. laboratory results), and clinical indication notes of the treating physician.⁸ The treating physician’s notes are of varying detail levels – they may provide no clinical information or guidance, request the analysis of a specific pathology, or

⁸Radiologists are rarely in direct contact with the patient or the treating physician, except via the formal information exchange outlined here.

only list the patient’s primary symptom (see appendix B.2 for examples). Irrespective of the pathology suspected by the treating physicians, radiologists are expected to report all pathological findings.

Because image-based classification is a core task performed by radiologists, a high-paying profession, it is not surprising that AI tools have made significant inroads in the field in the last decade. Recent advances in deep learning methods for image recognition have yielded algorithms that can match or surpass the performance of human radiologists (Obermeyer and Emanuel, 2016; Langlotz, 2019). As of 2020, 55 companies offered a total of 119 algorithmic products of which 46 have FDA approval (Tadavarthi et al., 2020). Most products related to clinical decision-making are marketed as support tools as opposed to autonomous tools, partly due to regulatory and liability issues (Harvey and Gowda, 2020).

3.1.2 *CheXpert*

We provide AI assistance using the CheXpert model, which is a deep learning prediction algorithm for chest X-ray interpretation (Irvin et al., 2019). This model is trained on a dataset of 224,316 chest radiographs of 65,240 patients labeled for the presence of fourteen common chest radiographic pathologies.⁹ The algorithm does not use any other patient information, such as the clinical history or vitals.¹⁰ Nonetheless, a prior version of this algorithm was shown to match or surpass the performance of board-certified radiologists from Stanford Hospital on five pathologies (Patel et al., 2019). These study results are also presented to our participants when introducing the AI tool. Section 4 confirms that the algorithm outperforms a majority of radiologists in our experiment. We relegate additional details about the algorithm to appendix B.3. The algorithm assistance to our participants will be in the form of a vector of probabilities for the presence of every pathology.¹¹

⁹The term artificial intelligence is typically reserved for a system of different prediction tasks to mimic a more complex set of behaviors, whereas machine learning is concerned with one specific prediction task. For a detailed discussion of this distinction see, for instance, (Taddy, 2018).

¹⁰While large datasets of images are increasingly available (e.g. Kramer et al. 2011, Johnson et al. 2016, Irvin et al. 2019) it is significantly more difficult to construct such datasets for other patient information due to the compulsory manual review of textual data for HIPPA compliance.

¹¹Some algorithms attempt to make their predictions explainable to a human by highlighting the parts of the image that drive a specific prediction. However, prior studies show that providing such localization in addition to the numeric output does not improve the accuracy of radiologists (Gaube et al., 2022). A quantitative output allows us to compute a Bayesian benchmark to the radiologist’s prediction, which is otherwise difficult.

3.2 Experimental Designs

Our experiment varies the information available to diagnose patient cases—participants may or may not receive AI assistance and access to the clinical history. The X-ray is shown under all information conditions. We expose our participants to all four possible information conditions: X-ray only, henceforth *XO*; clinical history without AI, henceforth *CH*; AI without clinical history, henceforth *AI*; and both clinical history and AI, henceforth *AI+CH*.

There are two objectives of our experiment. The first is to compute the treatment effects of AI and CH on diagnostic quality and radiologist time. The second is to analyze systematic deviations from the Bayesian benchmark.

Both these objectives are complicated by the likely heterogeneity in radiologist skills. For estimating treatment effects, radiologist heterogeneity implies that a design that randomizes treatments only across radiologists will require a large participant pool except for extremely large effect sizes. Our participants are highly paid experts, making this approach expensive. And, as explained in section 2, across-radiologist variation in information treatments is not tailored for the second objective. We would ideally know how a given radiologist changes her assessment for the same case under a different information condition.

Our approach to address these challenges is to use a combination of three different experimental designs, each with certain advantages and disadvantages. Appendix B.1 illustrates the three design variations.

3.2.1 Design 1 (Figure B.1)

In the first design, participants are assigned to a random sequence of the four information treatments. Each information condition is assigned fifteen cases at random without repetition. Participants read all 15 cases in one information environment before moving to the next one.

This design builds in both across- and within-participant variation in information treatments. The within-participant variation has greater power because it controls for participant heterogeneity at the potential cost of order effects. The concern of order effects is both testable and mitigated by the randomization of treatment sequence across subjects.

This first design is well-suited to estimate treatment effects of our information environments. However, as mentioned earlier, it is not ideal for estimating an empirical analog to equation (5) because no case is encountered twice.